



University of Vavuniya
Second Examination in Information Technology - 2022
First Semester - January/February 2024
IT2114 Data Structures (Theory)
Answer any Four Questions

Time Allowed : Two hours

1. (a) State *three* instances for each of the following data structures:
 - * i. *Linear data structures*
 - ↓ ii. *Non linear data structures*

[15%]
- (b) State the differences between an *algorithm*, a *pseudocode*, and a *program*. [15%]
- (c) Explain how the *selection sort method* works to sort the elements of an array [5, 2, 4, 6, 1, 3] show the step by step progress of each of the passes executed. [25%]
- (d) Write the *binary search* method in Java to find an element in a sorted array. [15%]
- (e) Trace how the method mentioned above in part 1(d) is used to search the target value 8 in the array [2, 5, 6, 8, 9, 10]. [15%]
- (f) Explain briefly why the binary search method is faster than the linear search method. [15%]

2. (a) Define each of the following data structures with its limitations:

i. An Array

ii. An ArrayList

iii. A Node

iv. A Linked List

[30%]

(b) Write a Java Class to define a Node for a single-linked list.

[20%]

(c) Write a Java class to define a Node for a double-linked list.

[15%]

(d) Consider the following double-linked list shown in Figure 1:

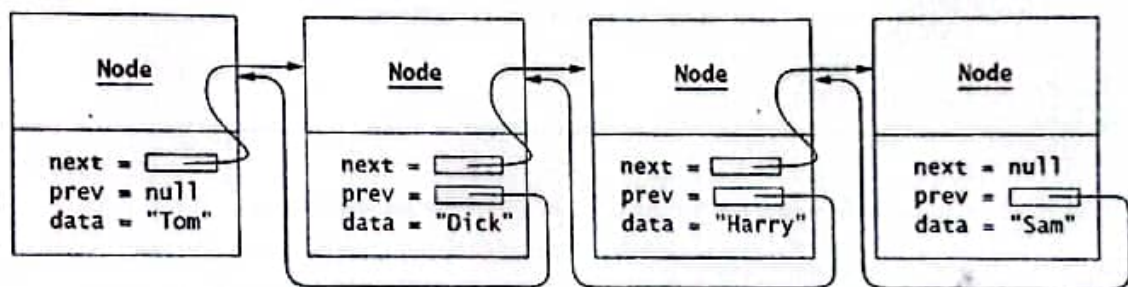


Figure 1: Double Linked List

Write the steps to insert a new node containing the data "Sharon" into the list immediately before the node with the data "Sam".

[15%]

(e) Write the steps to remove a node "Harry" from the double-linked List.

[15%]

(f) State the usage of circular-linked list in operating systems.

[05%]

3. (a) Write each of the following methods in a *Stack* abstract data type:
push(), *pop()*, *size()*, *isEmpty()*, *isFull()*, and *peek()*. [30%]

(b) Write each of the following methods in a *Queue* data structure:
enqueue() and *dequeue()*. [10%]

(c) Consider the following code snippet written in Java where the variables *stack* and *queue* refer a stack of integer objects and a queue of integer objects respectively.

Write the output of the following code:

```
for(int i = 0; i < 6; i++){
    stack.push(i);
for(int i = 0; i < 6; i++)
    queue.enqueue(stack.pop());
while(queue.size() > 2)
{
    System.out.println(queue.dequeue());
    queue.enqueue(queue.dequeue());
    queue.enqueue(queue.dequeue());
}
```

[25%]

(d) Explain briefly why it is important to handle *exceptions* in programming. [10%]

(e) Specify how each of the following *exceptions* can be thrown in programming:

i. *EmptyStackException*

ii. *StackOverflowError*

iii. *IndexOutOfBoundsException*

[25%]

4. (a) Define *trees* in the context of data structures. [10%]

(b) Define each of the following *binary tree* types with the aid of an example:

i. *Full binary tree*

ii. *Complete binary tree*

iii. *Perfect binary tree*

[15%]

(c) Construct a *binary search tree* for the elements [45, 10, 7, 90, 12, 50, 13, 39, 57]. [20%]

(d) i. Write an algorithm to delete a node from the binary search tree. [15%]

ii. Trace your algorithm when removing each of the nodes with the following values individually from the tree constructed in part 4(c).

A. 57

B. 90

C. 45

[20%]

(e) Determine the *inorder*, *preorder*, and *postorder* traversals of nodes in the tree shown in Figure 2 assuming that visiting a node displays the stored alphabet into the node:

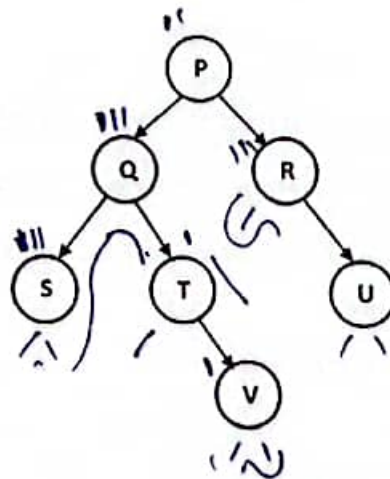


Figure 2: Tree

[20%]

5. (a) Consider the following graphs G1 and G2 given in Figure 3:

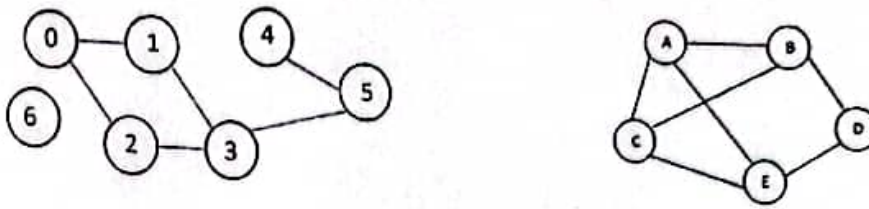


Figure 3: Graphs G1 & G2

- i. Construct the adjacency *matrix* representation for the graph G1. [20%]
 - ii. Construct the adjacency *list* representation for the graph G2. [20%]
- (b) Write an algorithm for each of the following types of traversals in graphs:
- i. *Depth first search*
 - ii. *Breadth first search*
- [30%]
- (c) Consider the graph G3 shown in Figure 4:

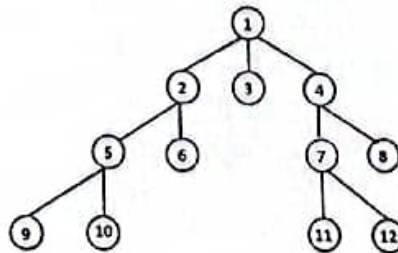


Figure 4: Graph G3

Determine the sequence of orders of vertices to be visited using each of the following searching methods from the vertex 1:

- i. *Depth first search* [10%]
 - ii. *Breadth first search* [10%]
- (d) State the importance of *Graph* data structure in computing. [10%]